

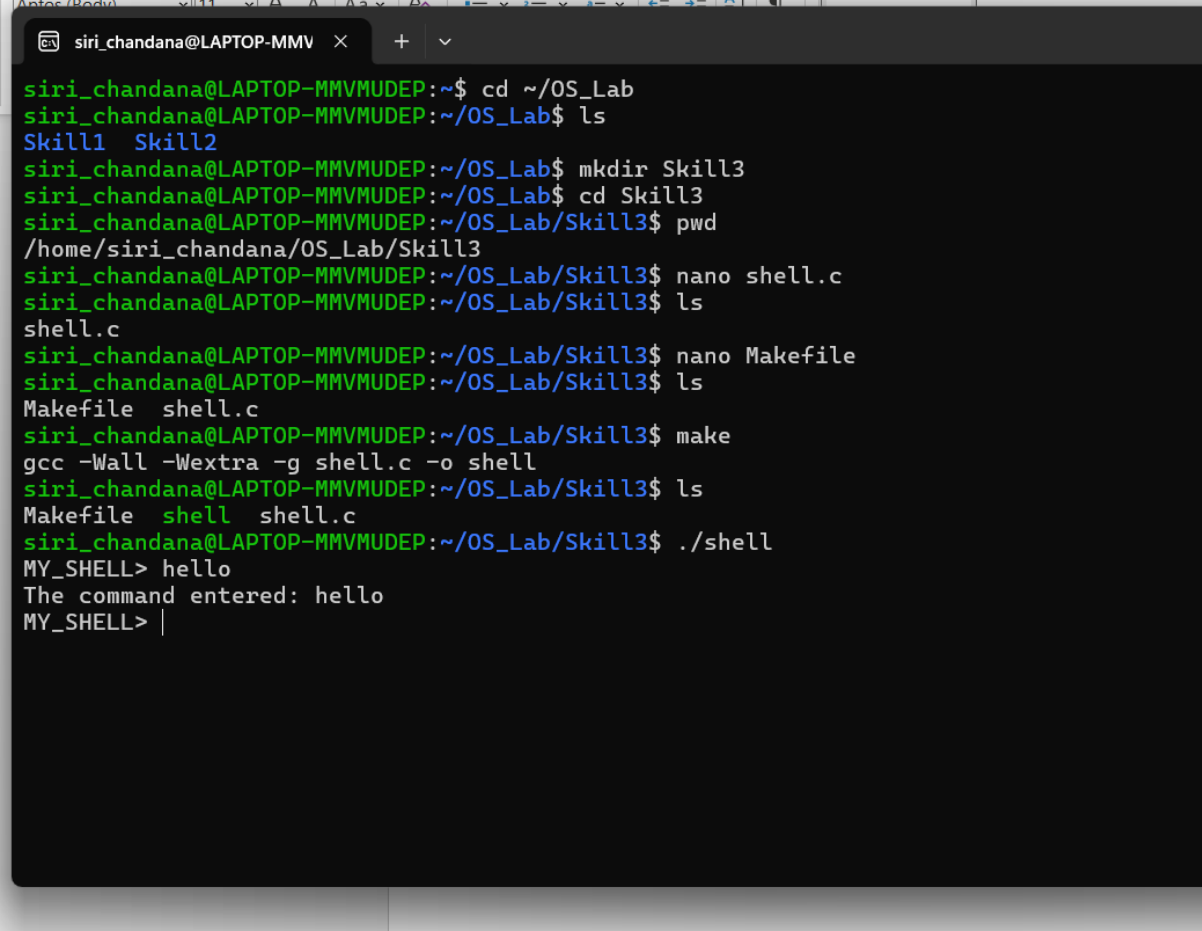
2520030453

K. Siri Chandana

Section-6 OS

Skill Programs

3.



```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUDEP:~$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ ls
Skill1 Skill2
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ mkdir Skill3
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ cd Skill3
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ pwd
/home/siri_chandana/OS_Lab/Skill3
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ nano shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ ls
shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ ls
Makefile shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ ls
Makefile shell shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ ./shell
MY_SHELL> hello
The command entered: hello
MY_SHELL> |
```

4.

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 parser.c *
    printf("\nToken stream is valid.\n");

    Node *tree = build_parse_tree(tokens, count);

    printf("\nParse Tree:\n");
    display_tree(tree, 0);

    free_tree(tree);

    printf("\n");
}

return 0;
}|
```

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 Makefile *
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = parser

all: $(TARGET)

$(TARGET): parser.c
    $(CC) $(CFLAGS) parser.c -o $(TARGET)

clean:
    rm -f $(TARGET)|
```

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill3$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ ls
Skill1 Skill2 Skill3
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ mkdir Skill4
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ cd Skill4
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill4$ pwd
/home/siri_chandana/OS_Lab/Skill4
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill4$ nano parser.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill4$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill4$ make
gcc -Wall -Wextra -g parser.c -o parser
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill4$ ls
Makefile parser parser.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill4$ ./parser
SKILL 4 - TOKENIZER AND PARSER
Type 'exit' to quit.

PARSER> ls -l

Tokens:
Token 1: [ls]
Token 2: [-l]

Token stream is valid.

Parse Tree:
|-- ls
  |-- -l

PARSER> |
```

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
SKILL 4 - TOKENIZER AND PARSER
Type 'exit' to quit.

PARSER> ls -l

Tokens:
Token 1: [ls]
Token 2: [-l]

Token stream is valid.

Parse Tree:
|-- ls
  |-- -l

PARSER> echo hello world

Tokens:
Token 1: [echo]
Token 2: [hello]
Token 3: [world]

Token stream is valid.

Parse Tree:
|-- echo
  |-- hello
    |-- world

PARSER> |
```

5.

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 quotes.c *
printf("QUOTES> ");
fflush(stdout);

if (fgets(input, sizeof(input), stdin) == NULL)
    break;

input[strcspn(input, "\n")] = '\0';

if (strcmp(input, "exit") == 0)
    break;

if (strlen(input) == 0)
{
    printf("Empty command.\n");
    continue;
}

Token tokens[MAX_TOKENS];

int count = tokenize(input, tokens);

if (count < 0)
    continue;

display_tokens(tokens, count);
```

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 Makefile *
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = quotes

all: $(TARGET)

$(TARGET): quotes.c
    $(CC) $(CFLAGS) quotes.c -o $(TARGET)

clean:
    rm -f $(TARGET)
```

Help **Write Out** **Where Is** **Cut** **Execute** **Location** **Undo** **Set Mark**
Exit **Read File** **Replace** **Paste** **Justify** **Go To Line** **Redo** **Copy**

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ ls
Skill1 Skill2 Skill3 Skill4
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ mkdir Skill5
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ cd Skill5
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill5$ pwd
/home/siri_chandana/OS_Lab/Skill5
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill5$ nano quotes.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill5$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill5$ make
gcc -Wall -Wextra -g quotes.c -o quotes
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill5$ ls
Makefile quotes quotes.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill5$ ./quotes
SKILL 5 - QUOTE HANDLING
Type 'exit' to quit.

QUOTES> 'hello world'

Tokens:
Token 1: [hello world]

Command parsed successfully.

QUOTES> |
```

```
Command parsed successfully.

QUOTES> '$HOME'

Tokens:
Token 1: [$HOME]

Command parsed successfully.

QUOTES> |
```

```
siri_chandana@LAPTOP-MIMV X 4 | v
QUOTES> '$HOME'
Tokens:
Token 1: [$HOME]
Command parsed successfully.
QUOTES> "hello world"
Tokens:
Token 1: [hello world]
Command parsed successfully.
QUOTES> "$HOME"
Tokens:
Token 1: [/home/siri_chandana]
Command parsed successfully.
QUOTES> "Hello $HOME"
Tokens:
Token 1: [Hello /home/siri_chandana]
Command parsed successfully.
QUOTES> |
```

6.

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 shell.c *
*
* Example:
* hello\ world
*
* becomes:
* hello world
*/
process_escapes(input);

char *args[MAX_ARGS];

int argc = parse_arguments(input, args);

if (argc <= 0 || args[0] == NULL)
    continue;

printf("[Shell] Command: %s\n", args[0]);

execute_command(args);

printf("\n");
}

return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo
```

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 Makefile *
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = shell

all: $(TARGET)

$(TARGET): shell.c
    $(CC) $(CFLAGS) shell.c -o $(TARGET)

clean:
    rm -f $(TARGET)
```

```

siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab$ cd Skill6
siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill6$ pwd
/home/siri_chandana/OS_Lab/Skill6
siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill6$ ls
Makefile  shell  shell.c
siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill6$ ./shell
SKILL 6 - PROCESS EXECUTION
Type 'exit' to quit.

EXEC> ls
[Shell] Command: ls
[Parent] PID: 1351, Child PID: 1358
[Child] PID: 1358
Makefile  shell  shell.c
[Parent] Child process completed.

EXEC> ls -l
[Shell] Command: ls
[Parent] PID: 1351, Child PID: 1367
[Child] PID: 1367
total 32
-rw-r--r-- 1 siri_chandana siri_chandana 150 Sep 17 09:40 Makefile
-rwxr-xr-x 1 siri_chandana siri_chandana 21200 Sep 17 09:40 shell
-rw-r--r-- 1 siri_chandana siri_chandana 2999 Sep 17 09:39 shell.c
[Parent] Child process completed.

EXEC> echo hello\ world
[Shell] Command: echo
[Parent] PID: 1351, Child PID: 1370
[Child] PID: 1370
hello world
[Parent] Child process completed.

EXEC> abcd123
[Shell] Command: abcd123
[Parent] PID: 1351, Child PID: 1371
[Child] PID: 1371
execvp: No such file or directory
[Parent] Child process completed.

EXEC> |

```

7.

```
siri_chandana@LAPTOP-MMV × + v
GNU nano 8.7.1 shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <sys/stat.h>

#define MAX_INPUT 1000
#define MAX_ARGS 100
#define MAX_PATH_LENGTH 4096

/*
 * Find an executable command by searching
 * every directory stored in PATH.
 */
char *find_executable(const char *command)
{
    char *path = getenv("PATH");

    if (path == NULL)
    {
        printf("PATH variable not found.\n");
        return NULL;
    }

    /* Make a copy because strtok modifies the string */
    char *path_copy = malloc(strlen(path) + 1);

    if (path_copy == NULL)
    {
        perror("malloc");
        return NULL;
    }

    strcpy(path_copy, path);

    char *directory = strtok(path_copy, ":");

    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E
```

```
siri_chandana@LAPTOP-MMV × + v
GNU nano 8.7.1 Makefile *
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = shell

all: $(TARGET)

$(TARGET): shell.c
    $(CC) $(CFLAGS) shell.c -o $(TARGET)

clean:
    rm -f $(TARGET)|
```

```
siri_chandana@LAPTOP-MMV X + v
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill7$ nano shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill7$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill7$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill7$ ls
Makefile  shell  shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill7$ ./shell
SKILL 7 - WAITPID AND PATH RESOLUTION
Type 'exit' to quit.

PATH_SHELL> ls
Executable found: /usr/bin/ls
[Parent] PID = 1555
[Parent] Child PID = 1556
[Child] PID = 1556
Makefile  shell  shell.c
[Parent] Child exited with status 0

PATH_SHELL> ls -l
Executable found: /usr/bin/ls
[Parent] PID = 1555
[Parent] Child PID = 1557
[Child] PID = 1557
total 32
-rw-r--r-- 1 siri_chandana siri_chandana 150 Sep 17 09:50 Makefile
-rwxr-xr-x 1 siri_chandana siri_chandana 21600 Sep 17 09:50 shell
-rw-r--r-- 1 siri_chandana siri_chandana 3970 Sep 17 09:49 shell.c
[Parent] Child exited with status 0

PATH_SHELL> pwd
Executable found: /usr/bin/pwd
[Parent] PID = 1555
[Parent] Child PID = 1558
[Child] PID = 1558
/home/siri_chandana/OS_Lab/Skill7
[Parent] Child exited with status 0

PATH_SHELL> abcdef123
abcdef123: command not found
```

8.

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 shell.c *
| fflush(stdout);

    if (fgets(input, sizeof(input), stdin) == NULL)
        break;

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0)
        continue;

    /* Expand variables */
    char expanded[MAX_VALUE];

    expand_variables(input, expanded);

    /* Copy expanded command because strtok modifies it */
    char command[MAX_VALUE];

    strcpy(command, expanded);

    char *args[MAX_ARGS];

    int argc = parse_arguments(command, args);

    if (argc == 0)
        continue;

    /* Exit */
    if (strcmp(args[0], "exit") == 0)
    {
        execute_builtin(args);
        break;
    }

    /* Built-in dispatch */
    if (execute_builtin(args))
```

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 Makefile *
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = shell

all: $(TARGET)

$(TARGET): shell.c
    $(CC) $(CFLAGS) shell.c -o $(TARGET)

clean:
    rm -f $(TARGET)|
```

```
siri_chandana@LAPTOP-MMV x + v
-rw-r--r-- 1 siri_chandana siri_chandana 150 Sep 17 09:50 Makefile
-rwxr-xr-x 1 siri_chandana siri_chandana 21600 Sep 17 09:50 shell
-rw-r--r-- 1 siri_chandana siri_chandana 3970 Sep 17 09:49 shell.c
[Parent] Child exited with status 0

PATH_SHELL> pwd
Executable found: /usr/bin/pwd
[Parent] PID = 1555
[Parent] Child PID = 1558
[Child] PID = 1558
/home/siri_chandana/OS_Lab/Skill7
[Parent] Child exited with status 0

PATH_SHELL> abcdef123
abcdef123: command not found

PATH_SHELL> exit
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill7$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ ls
Skill1 Skill2 Skill3 Skill4 Skill5 Skill6 Skill7
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ mkdir Skill8
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ cd Skill8
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill8$ pwd
/home/siri_chandana/OS_Lab/Skill8
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill8$ nano shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill8$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill8$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill8$ ls
Makefile shell shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill8$ ./shell
SKILL 8 - VARIABLES AND BUILT-INS
Type 'help' for commands.

VAR_SHELL> set name Siri
Variable name set to Siri
VAR_SHELL> get name
name = Siri
VAR_SHELL> echo $name
Siri
VAR_SHELL> |
```

9.



```
GNU nano 8.7.1 shell.c *
    args[count++] = token;

    token = strtok(NULL, " \\t");
}

args[count] = NULL;

return count;
}

int main()
{
    char input[MAX_INPUT];

    printf("SKILL 9 - DIRECTORY NAVIGATION\n");
    printf("Commands: cd, pwd, exit\n\n");

    while (1)
    {
        printf("CD_SHELL> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
            continue;

        char *args[MAX_ARGS];

        int argc = parse_arguments(input, args);

        if (argc == 0)
            continue;
```

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ cd Skill9
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ pwd
/home/siri_chandana/OS_Lab/Skill9
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ nano shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ make
make: *** No targets specified and no makefile found. Stop.
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ ls
Makefile shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ make
gcc -Wall -Wextra -g shell.c -o shell
shell.c:8:9: warning: 'MAX_INPUT' redefined
    8 | #define MAX_INPUT 1000
      | ^~~~~~
In file included from /usr/include/x86_64-linux-gnu/bits/local_lim.h:38,
                 from /usr/include/x86_64-linux-gnu/bits/posix1_lim.h:161,
                 from /usr/include/limits.h:198,
                 from /usr/lib/gcc/x86_64-linux-gnu/15/include/limits.h:210,
                 from /usr/lib/gcc/x86_64-linux-gnu/15/include/syslimits.h:
                 from /usr/lib/gcc/x86_64-linux-gnu/15/include/limits.h:34,
                 from shell.c:5:
/usr/include/linux/limits.h:11:9: note: this is the location of the previous
definition
   11 | #define MAX_INPUT      255      /* size of the type-ahead buffer */
      | ^~~~~~
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ sed -i 's/MAX_INPUT/INPUT_SIZE/g' shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ make clean
rm -f shell
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ ./shell
SKILL 9 - DIRECTORY NAVIGATION
Commands: cd, pwd, exit

CD_SHELL> pwd
/home/siri_chandana/OS_Lab/Skill9
CD_SHELL> cd -
Previous directory not available.
CD_SHELL> exit
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill9$ |
```

10.

```
siri_chandana@LAPTOP-MMV × + v
GNU nano 8.7.1 shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define INPUT_SIZE 1000
#define MAX_ARGS 100
#define MAX_VARS 100

struct Variable
{
    char name[100];
    char value[500];
};

struct Variable variables[MAX_VARS];
int variable_count = 0;

void cleanup()
{
    printf("\nCleaning up shell...\n");
    printf("Shell state saved. Exiting.\n");
}

void builtin_pwd()
{
    char current_directory[INPUT_SIZE];

    if (getcwd(current_directory, sizeof(current_directory)) == NULL)
    {
        perror("pwd");
        return;
    }

    printf("%s\n", current_directory);
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
siri_chandana@LAPTOP-MMV x + v - □ ×
;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:
*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=
01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;3
6:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.
ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xsp
f=00;36:*.~=00;90:.*#=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:
*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf
-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.
rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=siri_chandana
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/napd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games
:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/Microsof
tCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/P
rogram Files/Python310/Scripts:/mnt/c/Program Files/Python310:/mnt/c/Progr
am Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common
Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/
Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32
/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System3
2/OpenSSH:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/Program Files/nodejs/
:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/Siri Chandana/AppData/Local/Progr
ams/Python/Launcher:/mnt/c/Users/Siri Chandana/AppData/Local/Microsoft/Wind
owsApps:/mnt/c/Users/Siri Chandana/AppData/Local/Programs/Microsoft VS Code/
bin:/mnt/c/Users/Siri Chandana/AppData/Roaming/npm:/mnt/c/Users/Siri Chandan
a/AppData/Local/GitHubDesktop/bin:/mnt/c/Program Files/Python310:/mnt/c/Use
rs/Siri Chandana/AppData/Roaming/Python/Python310/Scripts:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
OLDPWD=/home/siri_chandana/OS_Lab
_=./shell
MYNAME=Chandana
ENV SHELL> |
```

11.

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define INPUT_SIZE 1000
#define HISTORY_CAPACITY 10
#define MAX_PIPE_COMMANDS 20

char *history[HISTORY_CAPACITY];
int history_count = 0;

/* Store a command in history */
void add_history(const char *command)
{
    if (strlen(command) == 0)
        return;

    if (history_count < HISTORY_CAPACITY)
    {
        history[history_count] = malloc(strlen(command) + 1);

        if (history[history_count] == NULL)
        {
            perror("malloc");
            return;
        }

        strcpy(history[history_count], command);
        history_count++;
    }
    else
    {
        /* Remove oldest command */
        free(history[0]);

        for (int i = 1; i < HISTORY_CAPACITY; i++)
        {

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

```

PULSE_SERVER=unix:/mnt/wslg/PulseServer
OLDPWD=/home/siri_chandana/OS_Lab
_=./shell
MYNAME=Chandana
ENV_SHELL> exit

Cleaning up shell...
Shell state saved. Exiting.
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill10$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ mkdir Skill11
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill11
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill11$ pwd
/home/siri_chandana/OS_Lab/Skill11
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill11$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill11$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill11$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill11$ ./shell
SKILL 11 - HISTORY AND PIPELINE STRUCTURES
Commands: history, !number, pipeline, exit

HISTORY_SHELL> pwd
Stored: pwd
HISTORY_SHELL> ls
Stored: ls
HISTORY_SHELL> date
Stored: date
HISTORY_SHELL> history
1  pwd
2  ls
3  date
HISTORY_SHELL> |

```

12.

```
siri_chandana@LAPTOP-MMV  ×  +  ▾
GNU nano 8.7.1                shell.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define INPUT_SIZE 1000
#define MAX_COMMANDS 20
#define MAX_ARGS 100

struct Pipeline
{
    char *commands[MAX_COMMANDS];
    int command_count;
};

void parse_pipeline(char *input, struct Pipeline *pipeline)
{
    pipeline->command_count = 0;

    char *command = strtok(input, "|");

    while (command != NULL &&
           pipeline->command_count < MAX_COMMANDS)
    {
        while (*command == ' ' || *command == '\t')
            command++;

        pipeline->commands[pipeline->command_count++] = command;

        command = strtok(NULL, "|");
    }
}

int parse_arguments(char *command, char *args[])
{
    int count = 0;

    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```

HISTORY_SHELL> ls
Stored: ls
HISTORY_SHELL> date
Stored: date
HISTORY_SHELL> history
1  pwd
2  ls
3  date
HISTORY_SHELL> exit
History memory released.
Exiting Skill 11.
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill11$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ mkdir Skill12
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill12
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ pwd
/home/siri_chandana/OS_Lab/Skill12
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ ./shell
SKILL 12 - PIPES AND FILE DESCRIPTORS
Example: ls | wc -l
Multiple pipes: ls | grep .c | wc -l

PIPE_SHELL> exit
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ |

```

13.

```
siri_chandana@LAPTOP-MMV x + v - □ ×
GNU nano 8.7.1 shell.c *

    execvp(args[0], args);

    perror(args[0]);
    exit(EXIT_FAILURE);
}

waitpid(pid, NULL, 0);
}

int main()
{
    char input[INPUT_SIZE];

    printf("SKILL 13 - INPUT/OUTPUT REDIRECTION\n");
    printf("Examples:\n");
    printf("  ls > output.txt\n");
    printf("  cat < output.txt\n");
    printf("  echo hello >> output.txt\n\n");

    while (1)
    {
        printf("REDIRECT_SHELL> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
            continue;

        if (strcmp(input, "exit") == 0)
            break;

        execute_command(input);
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify

```

siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ ./shell
SKILL 12 - PIPES AND FILE DESCRIPTORS
Example: ls | wc -l
Multiple pipes: ls | grep .c | wc -l

PIPE_SHELL> exit
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill12$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ mkdir Skill13
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill13
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill13$ pwd
/home/siri_chandana/OS_Lab/Skill13
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill13$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill13$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill13$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill13$ ./shell
SKILL 13 - INPUT/OUTPUT REDIRECTION
Examples:
  ls > output.txt
  cat < output.txt
  echo hello >> output.txt

REDIRECT_SHELL> echo Hello World > output.txt
REDIRECT_SHELL> cat output.txt
Hello World
REDIRECT_SHELL> cat < output.txt
Hello World
REDIRECT_SHELL> exit
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill13$ |

```

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 shell.c *

        if (fd < 0)
        {
            perror("error file");
            exit(EXIT_FAILURE);
        }

        dup2(fd, STDERR_FILENO);
        close(fd);
    }

    execvp(args[0], args);

    perror(args[0]);
    exit(EXIT_FAILURE);
}

waitpid(pid, NULL, 0);
}

int main()
{
    char input[INPUT_SIZE];

    printf("SKILL 14 - ERROR AND APPEND REDIRECTION\n");
    printf("Examples:\n");
    printf("  echo Hello > output.txt\n");
    printf("  echo Again >> output.txt\n");
    printf("  invalidcommand 2> errors.txt\n");
    printf("  invalidcommand 2>> errors.txt\n\n");

    while (1)
    {
        printf("REDIRECT_SHELL> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill13$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ mkdir Skill14
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ cd Skill14
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill14$ pwd
/home/siri_chandana/OS_Lab/Skill14
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill14$ nano shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill14$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill14$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill14$ ./shell
SKILL 14 - ERROR AND APPEND REDIRECTION
Examples:
  echo Hello > output.txt
  echo Again >> output.txt
  invalidcommand 2> errors.txt
  invalidcommand 2>> errors.txt

REDIRECT_SHELL> echo First > output.txt
REDIRECT_SHELL> echo Second >> output.txt
REDIRECT_SHELL> cat output.txt
First
Second
REDIRECT_SHELL> invalidcommand 2> errors.txt
REDIRECT_SHELL> cat errors.txt
invalidcommand: No such file or directory
REDIRECT_SHELL> anotherbadcommand 2>> errors.txt
REDIRECT_SHELL> cat errors.txt
invalidcommand: No such file or directory
anotherbadcommand: No such file or directory
REDIRECT_SHELL> exit
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill14$ |
```



```
GNU nano 8.7.1 shell.c *
|
|     flags |= O_APPEND;
|     else
|         flags |= O_TRUNC;
|
|     int fd = open(error_file, flags, 0644);
|
|     if (fd < 0)
|     {
|         perror("error file");
|         exit(EXIT_FAILURE);
|     }
|
|     if (dup2(fd, STDERR_FILENO) < 0)
|     {
|         perror("dup2");
|         close(fd);
|         exit(EXIT_FAILURE);
|     }
|
|     close(fd);
| }
|
| /*
|  * Merge stderr into stdout.
|  *
|  * This must happen after stdout has been
|  * redirected so both use the same destination.
|  */
| if (merge_output_error)
| {
|     if (dup2(STDOUT_FILENO, STDERR_FILENO) < 0)
|     {
|         perror("dup2");
|         exit(EXIT_FAILURE);
|     }
| }
|
```

 ^G Help	 ^O Write Out	 ^F Where Is	 ^K Cut	 ^T Execute
 ^X Exit	 ^R Read File	 ^\ Replace	 ^U Paste	 ^J Justify

```

invalidcommand: No such file or directory
REDIRECT_SHELL> anotherbadcommand 2>> errors.txt
REDIRECT_SHELL> cat errors.txt
invalidcommand: No such file or directory
anotherbadcommand: No such file or directory
REDIRECT_SHELL> exit
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill14$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ mkdir Skill15
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill15
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ pwd
/home/siri_chandana/OS_Lab/Skill15
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ cd ~/OS_Lab/Skill15
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ ./shell
SKILL 15 - COMBINED REDIRECTION
Examples:
    echo Hello > output.txt
    invalidcommand 2> errors.txt
    invalidcommand > output.txt 2>&1
    echo More >> output.txt

COMBINED_SHELL> echo Hello > output.txt
COMBINED_SHELL> cat output.txt
Hello
COMBINED_SHELL> exit
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ |

```



```
GNU nano 8.7.1 shell.c *

    int background = 0;

    int length = strlen(input);

    if (length > 0 && input[length - 1] == '&')
    {
        background = 1;

        input[length - 1] = '\\0';

        while (length > 1 &&
            (input[length - 2] == ' ' ||
             input[length - 2] == '\\t'))
        {
            input[length - 2] = '\\0';
            length--;
        }
    }

    execute_command(input, background);
}

/*
 * Wait for remaining background jobs
 * before exiting.
 */
for (int i = 0; i < job_count; i++)
{
    if (jobs[i].active)
    {
        waitpid(jobs[i].pid, NULL, 0);
    }
}

printf("All jobs completed. Exiting.\\n");
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^\\ Replace	^U Paste	^J Justify

```
COMBINED_SHELL> exit
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill15$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ mkdir Skill16
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill16
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill16$ pwd
/home/siri_chandana/OS_Lab/Skill16
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill16$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill16$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill16$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill16$ ./shell
SKILL 16 - BACKGROUND JOBS
Example: sleep 5 &
Use 'jobs' to check background jobs.

JOB_SHELL> sleep 10 &
[1] 1449 running in background: sleep 10
JOB_SHELL> jobs
[1] 1449 running  sleep 10
JOB_SHELL> exit
All jobs completed. Exiting.
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill16$ |
```

17.

```
    exit(EXIT_FAILURE);
}

if (background)
{
    add_job(pid, command_copy);
}
else
{
    waitpid(pid, NULL, 0);
}
}

int main()
{
    char input[INPUT_SIZE];

    printf("SKILL 17 - JOBS AND FOREGROUND\n");
    printf("Commands:\n");
    printf("  sleep 10 &\n");
    printf("  jobs\n");
    printf("  fg 1\n");
    printf("  exit\n\n");

    while (1)
    {
        check_jobs();

        printf("JOB_SHELL> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
```

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUDEP:~$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ mkdir Skill17
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab$ cd Skill17
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill17$ pwd
/home/siri_chandana/OS_Lab/Skill17
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill17$ nano shell.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill17$ nano Makefile
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill17$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill17$ ./shell
SKILL 17 - JOBS AND FOREGROUND
Commands:
    sleep 10 &
    jobs
    fg 1
    exit

JOB_SHELL> sleep 10 &
[1] 1614 running in background: sleep 10
JOB_SHELL> jobs
[1] 1614 running  sleep 10
JOB_SHELL> fg 1
Bringing job [1] to foreground: sleep 10
Job [1] completed.
JOB_SHELL> exit
All jobs completed. Exiting.
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill17$ |
```

18.

```
| if (strlen(input) == 0)
    continue;

    if (strcmp(input, "exit") == 0)
        break;

    if (strcmp(input, "jobs") == 0)
    {
        list_jobs();
        continue;
    }

    if (strncmp(input, "cont ", 5) == 0)
    {
        int job_id = atoi(input + 5);

        continue_job(job_id);
        continue;
    }

    int background = 0;

    int length = strlen(input);

    if (length > 0 && input[length - 1] == '&')
    {
        background = 1;

        input[length - 1] = '\\0';

        while (length > 1 &&
            (input[length - 2] == ' ' ||
             input[length - 2] == '\\t'))
        {
            input[length - 2] = '\\0';
            length--;
        }
    }
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```

siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill17$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ mkdir Skill18
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill18
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill18$ pwd
/home/siri_chandana/OS_Lab/Skill18
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill18$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill18$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill18$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill18$ ./shell
SKILL 18 - SIGNALS AND PROCESS GROUPS
Commands:
    sleep 20 &
    jobs
    cont 1
    exit

SIGNAL_SHELL> sleep 20 &
[1] PID=1715 PGID=1715 running: sleep 20
SIGNAL_SHELL> jobs
[1] PID=1715 PGID=1715 RUNNING  sleep 20
SIGNAL_SHELL> cont 1
SIGCONT sent to job [1], process group 1715.
SIGNAL_SHELL> exit
All jobs cleaned up. Exiting.
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill18$ |

```

19.

```
siri_chandana@LAPTOP-MMV × + v
GNU nano 8.7.1 shell.c *
}
}

int main()
{
    char input[INPUT_SIZE];

    /*
     * Install SIGINT handler for the shell.
     */
    struct sigaction sa;

    memset(&sa, 0, sizeof(sa));

    sa.sa_handler = handle_sigint;

    sigemptyset(&sa.sa_mask);

    sa.sa_flags = 0;

    if (sigaction(SIGINT, &sa, NULL) == -1)
    {
        perror("sigaction");
        return 1;
    }

    printf("SKILL 19 - SIGNAL HANDLERS AND SIGINT\n");
    printf("Press Ctrl+C to interrupt the current command.\n");
    printf("The shell itself will continue running.\n\n");

    while (1)
    {
        interrupted = 0;

        printf("SIGNAL_SHELL> ");
        fflush(stdout);
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill19
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill19$ pwd
/home/siri_chandana/OS_Lab/Skill19
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill19$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill19$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill19$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill19$ ./shell
SKILL 19 - SIGNAL HANDLERS AND SIGINT
Press Ctrl+C to interrupt the current command.
The shell itself will continue running.

SIGNAL_SHELL> ^C
SIGINT received. Shell is still running.
SIGNAL_SHELL> sleep 20
^C
SIGINT received. Shell is still running.
SIGNAL_SHELL> echo hello
hello
SIGNAL_SHELL> exit
Shell exiting normally.
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill19$
```

20.

siri_chandana@LAPTOP-MMV × + ▾

GNU nano 8.7.1shell.c *

```
{
    perror("setpgid");
}

/*
 * Shell owns the terminal.
 */
tcsetpgrp(shell_terminal, shell_pgid);

/*
 * Ignore job-control signals in the shell.
 */
signal(SIGTSTP, SIG_IGN);
signal(SIGTTIN, SIG_IGN);
signal(SIGTTOU, SIG_IGN);
signal(SIGINT, SIG_IGN);

printf("SKILL 20 - SIGTSTP AND TERMINAL CONTROL\n");
printf("Run a command and press Ctrl+Z to stop it.\n");
printf("Use 'jobs' to list stopped jobs.\n");
printf("Use 'fg 1' to resume job 1.\n\n");

while (1)
{
    printf("JOB_CONTROL_SHELL> ");
    fflush(stdout);

    if (fgets(input, sizeof(input), stdin) == NULL)
        break;

    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0)
        continue;

    if (strcmp(input, "exit") == 0)
        break;
}
```

^G Help

^X Exit

^O Write Out

^R Read File

^F Where Is

^_ Replace

^K Cut

^U Paste

^T Execute

^J Justify

```

siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab$ cd Skill20
siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill20$ nano shell.c
siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill20$ nano Makefile
siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill20$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill20$ ./shell
SKILL 20 - SIGTSTP AND TERMINAL CONTROL
Run a command and press Ctrl+Z to stop it.
Use 'jobs' to list stopped jobs.
Use 'fg 1' to resume job 1.

JOB_CONTROL_SHELL> sleep 30
^Z
Process stopped by signal 20.
[1] PID=1906 PGID=1906 running: sleep 30
JOB_CONTROL_SHELL> jobs
[1] PID=1906 PGID=1906 STOPPED  sleep 30
JOB_CONTROL_SHELL> fg 1
Job [1] continued in foreground.
exit
Job [1] completed.
JOB_CONTROL_SHELL> All jobs cleaned up. Exiting.

siri_chandana@LAPTOP-MMVNUDEP:~/OS_Lab/Skill20$

```

21.

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 shell.c *
|
    }

    continue;
}

if (strchr(input, '|') != NULL)
{
    execute_pipeline(input);
    continue;
}

int background = 0;

int length = strlen(input);

if (length > 0 && input[length - 1] == '&')
{
    background = 1;
    input[length - 1] = '\\0';

    while (length > 1 &&
           (input[length - 2] == ' ' ||
            input[length - 2] == '\\t'))
    {
        input[length - 2] = '\\0';
        length--;
    }

    execute_single_command(input, background);
}

return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\\ Replace   ^U Paste     ^J Justify
```

```
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ mkdir Skill21
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab$ cd Skill21
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill21$ pwd
/home/siri_chandana/OS_Lab/Skill21
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill21$ nano shell.c
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill21$ nano Makefile
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill21$ make
gcc -Wall -Wextra -g shell.c -o shell
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill21$ ./shell
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> pwd
/home/siri_chandana/OS_Lab/Skill21
OS_SHELL> cd /tmp
OS_SHELL> pwd
/tmp
OS_SHELL> ls | wc -l
3
OS_SHELL> echo Hello > test.txt
OS_SHELL> cat test.txt
Hello
OS_SHELL> exit
Cleanup completed.
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill21$ |
```

22.

```
siri_chandana@LAPTOP-MMV  ×  +  ∨  -  □  ×

Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up libc-dev-bin (2.43-2ubuntu2.4) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Setting up libc6-dev:amd64 (2.43-2ubuntu2.4) ...
Processing triggers for systemd (259.5-0ubuntu3) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
siri_chandana@LAPTOP-MMV$ valgrind --version
valgrind-3.26.0
siri_chandana@LAPTOP-MMV$ valgrind --leak-check=full -
-show-leak-kinds=all ./shell
==3230== Memcheck, a memory error detector
==3230== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==3230== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==3230== Command: ./shell
==3230==
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> pwd
/home/siri_chandana/OS_Lab/Skill21
OS_SHELL> echo test
test
OS_SHELL> exit
Cleanup completed.
==3230==
==3230== HEAP SUMMARY:
==3230==    in use at exit: 0 bytes in 0 blocks
==3230==   total heap usage: 2 allocs, 2 frees, 2,048 bytes allocated
==3230==
==3230== All heap blocks were freed -- no leaks are possible
==3230==
==3230== For lists of detected and suppressed errors, rerun with: -s
==3230== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
siri_chandana@LAPTOP-MMV$ |
```

23.

```
siri_chandana@LAPTOP-MMV × + ▾
GNU nano 8.7.1 stress_test.sh *

./shell <<EOF
sleep 1 &
jobs
exit
EOF

echo ""
echo "[5] Testing invalid commands..."

./shell <<EOF
invalid_command_test
exit
EOF

echo ""
echo "=====|
echo " STRESS TEST COMPLETED"
echo "=====|
```

```
siri_chandana@LAPTOP-MMV  ×  +  ∨
Supports pipes, redirection and background jobs.

OS_SHELL> 1
OS_SHELL> Cleanup completed.

[3] Testing output redirection...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> OS_SHELL> StressTest
OS_SHELL> Cleanup completed.

[4] Testing background jobs...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> [1] 3379 running: sleep 1
OS_SHELL> [1] 3379 running sleep 1
OS_SHELL> Cleanup completed.

[5] Testing invalid commands...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> Runtime error: command 'invalid_command_test' not found: No such f
ile or directory
OS_SHELL> Cleanup completed.

=====
STRESS TEST COMPLETED
=====
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill23$ nano README.md
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill23$ ./shell
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.
```

24.

```
siri_chandana@LAPTOP-MMV  ×  +  ▾  -  □  ×

Test command 19
Test command 20

[2] Testing pipes...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> 1
OS_SHELL> Cleanup completed.

[3] Testing output redirection...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> OS_SHELL> StressTest
OS_SHELL> Cleanup completed.

[4] Testing background jobs...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> [1] 3550 running: sleep 1
OS_SHELL> [1] 3550 running  sleep 1
OS_SHELL> Cleanup completed.

[5] Testing invalid commands...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> Runtime error: command 'invalid_command_test' not found: No such f
ile or directory
OS_SHELL> Cleanup completed.

=====
STRESS TEST COMPLETED
=====
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/Skill24$ |
```

```
siri_chandana@LAPTOP-MMV  x  +  v  -  □  ×

Supports pipes, redirection and background jobs.

OS_SHELL> [1] 3550 running: sleep 1
OS_SHELL> [1] 3550 running sleep 1
OS_SHELL> Cleanup completed.

[5] Testing invalid commands...
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> Runtime error: command 'invalid_command_test' not found: No such f
ile or directory
OS_SHELL> Cleanup completed.

=====
  STRESS TEST COMPLETED
=====
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill24$ ./shell
SKILL 21 - INTEGRATED SHELL
Commands: cd, pwd, jobs, exit
Supports pipes, redirection and background jobs.

OS_SHELL> pwd
/home/siri_chandana/OS_Lab/Skill24
OS_SHELL> echo Hello | wc -l
1
OS_SHELL> echo FinalTest > final.txt
OS_SHELL> cat final.txt
FinalTest
OS_SHELL> sleep 3 &
[1] 3575 running: sleep 3
OS_SHELL> jobs

[1] finished: sleep 3
OS_SHELL> exit
Cleanup completed.
siri_chandana@LAPTOP-MMVMUDEP:~/OS_Lab/Skill24$ ls
Makefile  final.txt  shell.c          stress_test.sh
README.md  shell     stress_output.txt
```